

# Motor Datasheet: tm SAR

## 20260127 165414 20C

Operating Point: Continuous | Temperature: 20 °C

### Geometry (active elements only)

| Parameter                       | Value | Unit |
|---------------------------------|-------|------|
| Machine Outer Diameter          | 55.0  | mm   |
| Machine Inner Diameter          | 20.0  | mm   |
| Machine Thickness / Width       | 24.2  | mm   |
| Air Gap                         | 1.00  | mm   |
| Fill Factor (input)             | 0.65  | -    |
| Slot Fill Factor (calculated)   | 0.58  | -    |
| Slot Width (tooth to tooth)     | 4.292 | mm   |
| Available Coil Space            | 2.146 | mm   |
| Assumed Coil Width (simulation) | 1.946 | mm   |

### Mass Properties (active elements only)

| Parameter                                      | Value | Unit              |
|------------------------------------------------|-------|-------------------|
| Machine Mass                                   | 271.6 | g                 |
| Rotor Mass                                     | 60.3  | g                 |
| Stator Mass                                    | 211.2 | g                 |
| Rotor Structure Included                       | No    | -                 |
| Rotor Moment of Inertia (active elements only) | 245.9 | g·cm <sup>2</sup> |

## Electrical Properties

| Parameter                                 | Value      | Unit  |
|-------------------------------------------|------------|-------|
| Supply Current (RMS)                      | 4.583      | A     |
| Supply Current (Peak)                     | 9.037      | A     |
| Id (Peak)                                 | -0.000     | A     |
| Iq (Peak)                                 | 9.037      | A     |
| Phase Advance Angle                       | 0.0        | °elec |
| Phase Resistance                          | 69.732     | mOhm  |
| D-Axis Inductance (Ld)                    | 0.000      | μH    |
| Q-Axis Inductance (Lq)                    | 0.000      | μH    |
| Terminal Inductance (Ltt = Ld + Lq)       | 0.000      | μH    |
| PM Flux Linkage                           | 0.000      | mWb   |
| Frequency at operating point (electrical) | 237.500    | Hz    |
| Winding Configuration                     | wye (star) | -     |
| DC Supply Voltage                         | 9.000      | V     |
| Power Factor                              | 0.837      | -     |

## Performance

| Parameter                                  | Value  | Unit      |
|--------------------------------------------|--------|-----------|
| Torque (Continuous)                        | 120.0  | mNm       |
| Speed                                      | 2850.0 | rpm       |
| BackEMF Constant (Ke, phase peak)          | 1.3    | mV/rpm    |
| Motor Constant (Km, only resistive losses) | 57.3   | mNm/√W    |
| Specific Motor Constant (Km / mass)        | 210.8  | mNm/√W/kg |
| Torque Constant (Kt, peak, no-load)        | 18.4   | mNm/A     |
| Torque Ripple                              | 2.3    | %         |

|            |     |   |
|------------|-----|---|
| Saturation | 0.0 | % |
|------------|-----|---|

## Losses

| Parameter                           | Value | Unit |
|-------------------------------------|-------|------|
| Total Losses                        | 4.8   | W    |
| Resistive Losses                    | 4.4   | W    |
| Iron Core Losses                    | 0.7   | W    |
| Magnet Eddy Current Losses          | 0.0   | W    |
| Rotor Structure Eddy Current Losses | 0.0   | W    |
| Winding Eddy Current Losses         | 0.0   | W    |

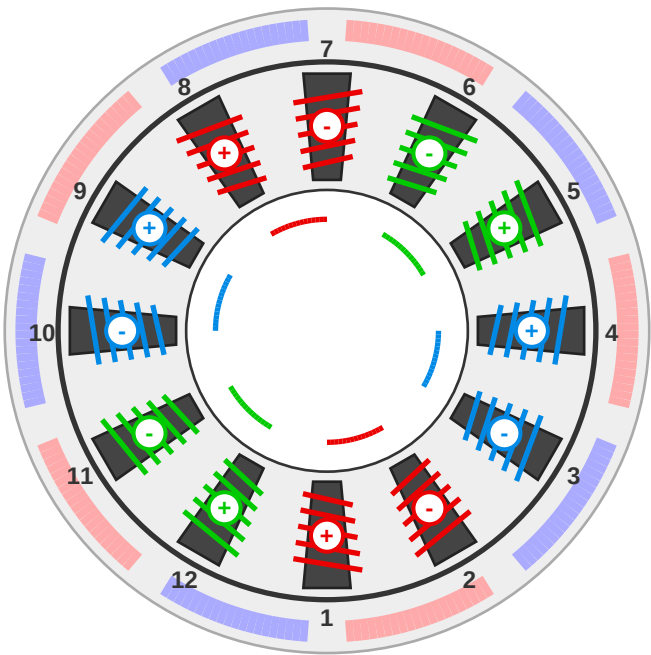
## Configuration

| Parameter                                                 | Value | Unit |
|-----------------------------------------------------------|-------|------|
| Number of Stators                                         | 1     | -    |
| Number of Rotors                                          | 1     | -    |
| Number of Slots                                           | 12    | -    |
| Pole Pairs                                                | 5     | -    |
| Number of Parallel Paths                                  | 1     | -    |
| Number of Turns per Tooth (designed incl. parallel paths) | 20.0  | -    |
| Number of Phase Turns (all phases in series)              | 80.0  | -    |

# Winding Configuration

Schema: **AabBCcaABbcC**

Concentrated Winding | Balanced | Cogging: 60/rev | Ripple: 60/rev



# Materials

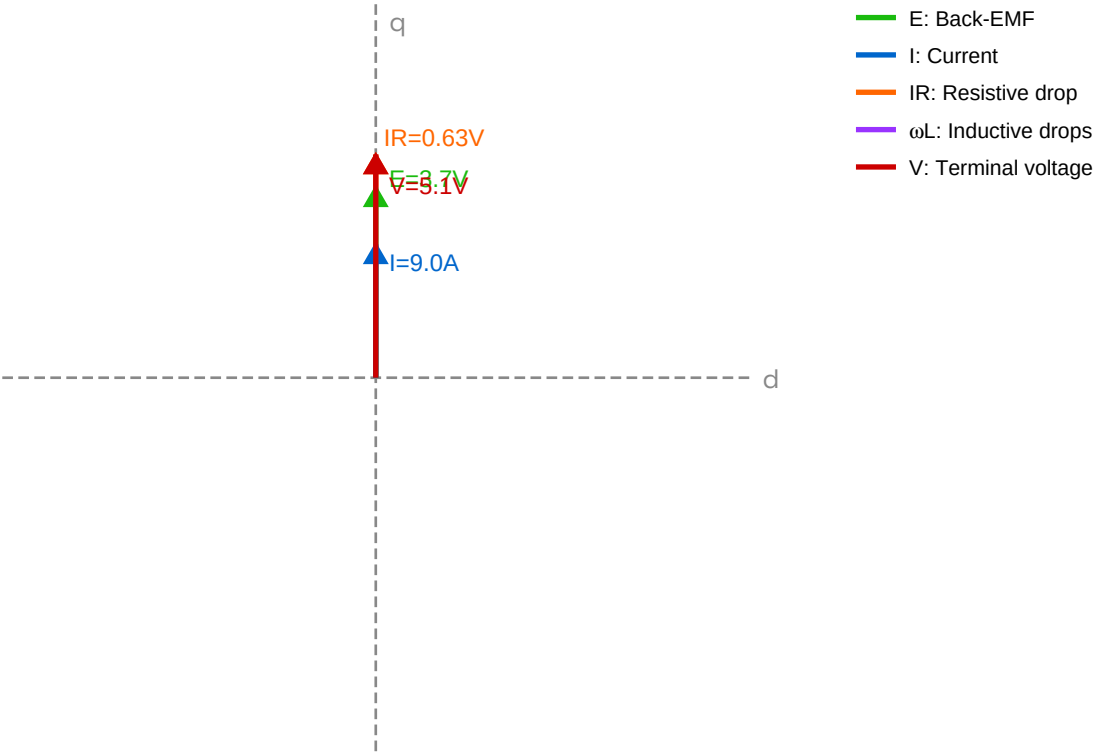
| Component      | Material                |   |
|----------------|-------------------------|---|
| Stator Core    | 700HR 5P 800MPa-Hoganas | - |
| Magnets        | BMHFa-32/27             | - |
| Rotor Backiron | 430-FR 15mm-Carpenter   | - |
| Coil           | Copper                  | - |

# Voltage Breakdown at Operating Point (with integer turns)

| Parameter                      | Value | Unit |
|--------------------------------|-------|------|
| Back-EMF Voltage (Phase, Peak) | 3.657 | V    |
| Resistive Drop (Phase, Peak)   | 0.630 | V    |
| Inductive Drop Ld (Peak)       | 0.000 | V    |
| Inductive Drop Lq (Peak)       | 0.000 | V    |

|                                       |       |   |
|---------------------------------------|-------|---|
| Inductive Drop Total (Peak)           | 0.000 | V |
| Terminal Voltage (Phase, Peak)        | 5.124 | V |
| Back-EMF Voltage (Line-to-Line, Peak) | 6.335 | V |
| Resistive Drop (Line-to-Line, Peak)   | 1.092 | V |
| Inductive Drop (Line-to-Line, Peak)   | 0.000 | V |
| Terminal Voltage (Line-to-Line, Peak) | 8.875 | V |

Phasor Diagram Phase Values NOT Line-to-Line (d-q reference frame)



## Mechanical Properties

| Parameter                           | Value  | Unit |
|-------------------------------------|--------|------|
| Airgap Axial Force (Average)        | 30.82  | N    |
| Magnet Axial Force Positive Maximum | 0.00   | N    |
| Magnet Axial Force Positive Minimum | 0.00   | N    |
| Magnet Axial Force Negative Maximum | -13.72 | N    |
| Magnet Axial Force Negative Minimum | -9.80  | N    |

## Glossary

- Peak — Refers to 0-to-peak amplitude (NOT peak-to-peak). For sinusoidal waveforms:  $\text{Peak} = \text{RMS} \times \sqrt{2}$ .
- RMS — Root Mean Square, the equivalent DC value for power calculations.
- Line-to-Line (L-L) — Voltage measured between two phases. For wye connection:  $V_{LL} = V_{\text{phase}} \times \sqrt{3}$ .